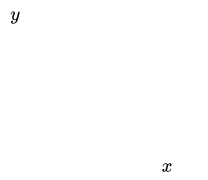
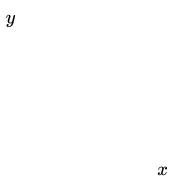
1.18 Graph Sketching: Rational Functions

Chapter 4 · Original slide 21 · GCSE RECAP: Reciprocal Graphs



Sketch $y = \frac{1}{x}$



Sketch $y = -\frac{3}{x}$

Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

Notice the distance between this line and the x-axis (i.e. the line $y = 0$) gradually decreases as the lines go off towards infinity. The line $y = 0$ is known as an asymptote of the graph.

! An asymptote is a line which the graph approaches but never reaches.

Asymptotes of $y = \frac{a}{x}$:

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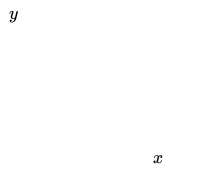
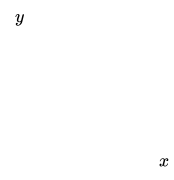


This is new to the A Level 2017 syllabus.



Sketch $y = \frac{3}{x^2}$

Sketch $y = -\frac{4}{x^2}$

Hint: Note that anything squared will always be at least 0.

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On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

y

$y = \frac{3}{x}$

The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

$y = \frac{1}{x}$

x

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1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$
d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

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2 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$